Kun Wang

Computational Biology Researcher | Stochastic Modeling & Single-Cell Lineage Dynamics

Shenzhen, Guangdong   kunwang034@gmail.com   kunwang34   Google Scholar

Introduction

PhD candidate in computational biology with training in probability theory and mathematical statistics. Specializes in stochastic dynamical modeling and statistical inference for phylogeny-resolved single-cell data. Developed computational frameworks integrating lineage tracing and transcriptomics to reconstruct developmental dynamics and infer cell fate decisions. Led the development of PhyloVelo (*Nature Biotechnology*, 2023) and scPhyloX (*Cell Systems*, 2025), with additional contributions to mitochondrial lineage tracing analysis (*Genome Biology*, 2025). First author of a review article accepted in *Nature Reviews Genetics*.

Education

PhD **Xiamen University**, Probability Theory and Mathematical Statistics — Xiamen, China
Sept 2022 – June 2026
- Thesis: Dynamical Modeling and Cell Fate Inference from Single-Cell Multimodal Lineage Tracing Data
- Advisor: Prof. Da Zhou

MS **Xiamen University**, Probability Theory and Mathematical Statistics — Xiamen, China
Sept 2019 – June 2022
- Advisor: Prof. Da Zhou

BS **Ocean University of China**, Mathematics and Applied Mathematics — Qingdao, China
Sept 2015 – June 2019

Experience

Shenzhen Institutes of Advanced Technology, Visiting Research Student — Shenzhen, China
Aug 2020 – June 2026
5 years 11 months
- Advisor: Dr. Zheng Hu
- Conducted research on lineage tracing and single-cell dynamics.
- Developed stochastic models integrating phylogeny and transcriptomics.

Research Contributions

PhyloVelo
June 2021 – June 2023

PhyloVelo infers cell state transitions using lineage-resolved single-cell data.
- Published in Nature Biotechnology (2023).
- Improves velocity estimation by incorporating lineage structure.
- Robustly maps developmental trajectories across diverse biological systems.

scPhyloX
Oct 2020 – Mar 2025

scPhyloX infers tissue growth dynamics from single-cell lineage trees
- Published in Cell Systems (2025).
- Quantifies diverse stem cell growth rates across different tissues.
- Detects Darwinian selection and expansion of advantageous tumor clones.

Skills

Programming: Python(Pytorch, Pyro, PyMC), R (Seurat), Matlab, Mathematica

Research Areas: Stochastic dynamical modeling, Variational inference, Single-cell lineage tracing, Transcriptomic analysis

Selected Publications

PhyloVelo Enhances Transcriptomic Velocity Field Mapping Using Monotonically Expressed Genes — May 2024
Kun Wang, Liangzhen Hou, et al.
10.1038/s41587-023-01887-5 (*Nature Biotechnology*)

Single-Cell Phylodynamic Inference of Stem Cell Differentiation and Tumor Evolution — May 2025
Kun Wang, Zhaolian Lu, et al.
10.1016/j.cels.2025.101244 (*Cell Systems*)

Computational approaches for multimodal lineage tracing — Apr 2026
Kun Wang, Xionglei He, Zheng Hu
(*Nature Reviews Genetics*, Accepted)

Clonal Expansion Dictates the Efficacy of Mitochondrial Lineage Tracing in Single Cells — Jan 2025
Xin Wang*, **Kun Wang***, et al.
10.1186/s13059-025-03540-7 (*Genome Biology*)

Polyclonal-to-Monoclonal Transition in Colorectal Precancerous Evolution — Jan 2024
Zhaolian Lu, Shanlan Mo, et al.
10.1038/s41586-024-08133-1 (*Nature*)

Selected Honors

- Top 10 Bioinformatics Advances in China, 2023
- Best Oral Presentation, GPB Omics & Bioinformatics Frontiers Symposium, 2023
- BYD Scholarship, 2026
- SIAT Dean's Scholarship, 2023

Invited Talks

1. GPB Omics & Bioinformatics Frontiers Symposium 2023: "PhyloVelo enhances transcriptomic velocity field mapping using monotonically expressed genes" (Best Oral Presentation)
2. 2023 Genomics & AI: First Author Forum: "PhyloVelo enhances transcriptomic velocity field mapping using monotonically expressed genes"
3. 2025 Genomics & AI: First Author Forum: "Single-cell phylodynamic inference of stem cell differentiation and tumor evolution"
4. 2025 Quantitative Biology: Mathematical Modeling and Statistical Inference Workshop: "Stochastic modeling of cell fate dynamics in lineage-traced single-cell data"